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What is velocity based training? a quick start guide to VBT

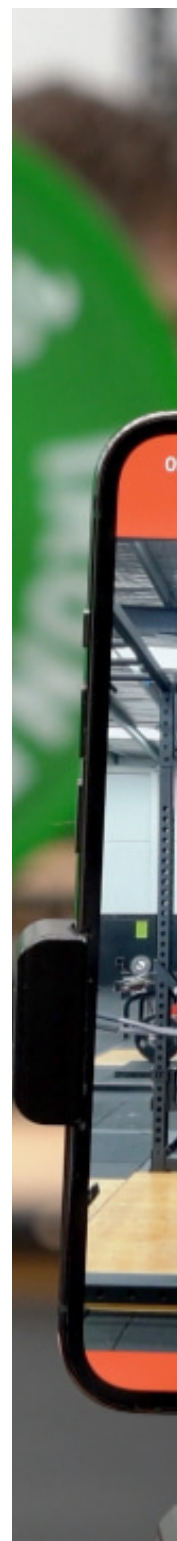
More than just lifting light weights fast, Velocity based training (VBT) opens up a world of programming and training possibilities



Jacob Tober
Co-founder

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Velocity tracking, also known as velocity based training (VBT), measures the movement and speed of your repetitions in weight



training. This data helps coaches and athletes make better training decisions and improve performance.

Velocity tracking is similar to how GPS data is used by runners to track and improve their training.

What is Velocity Based Training?

As a top-level summary, velocity based training (VBT) is a novel approach to strength and conditioning that takes into account an individual's movement speed in order to track progress and optimise performance. It is a data-driven approach that uses some form of sensors (camera, string, accelerometer) and software to measure bar velocity, force, power, and other key performance metrics. This information is then used by coaches and lifters to prescribe training intensity, volume, and more in real-time, providing athletes with immediate feedback on their performance.

The ultimate goal of doing this is that training can be more finely tuned to respond to the multiple factors that influence performance on a day-to-day basis. This includes, fatigue, readiness, motivation, peaking and tapering, injury, and faster or slower rates of adaptation.

Velocity based training measures quality in training

- Velocity indicates force and power output. Faster bar speed requires more force application against the load.

Faster velocities over time are an indication of improving strength and

power, while a drop in velocity between weeks might indicate fatigue accumulation or poor technique.

Use your velocity tracking to motivate intent in your training, track progress on more dimensions, and detect fatigue in real time.

There are dozens of ways to apply this information to your training and programming to improve strength, power or hypertrophy. Check out the programming tips for some ideas on how to use velocity in your training.

Velocity based training (VBT) is growing in popularity thanks to its unique benefits for athletic performance and strength gains. But what makes tracking velocity so useful?

The benefits of VBT

Barbell velocity that improves indicates increasing strength

An increase in lifting velocity at the same weight is an indication that your force and power production have increased. An increase in mean velocity of as little as 0.05m/s can indicate an increase in 1RM strength of 2.5%.

Velocity loss goes up with fatigue

This can be an early warning before a set reaches failure, injury or due to overtraining. Drops act as an early warning sign of fatigue and help coaches and lifters regulate their training sets.

Barbell velocity tracking is objective

It provides quantification of effort and progress beyond reps completed or weight lifted.

Velocity guides training load

Lifts can be regulated in real-time to match readiness on any given day.

[For more information on the uses for velocity based training check out this blog post →](#)

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